

Machine Learning for Business

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Key Points of the Course

Objective

- ▶ Understand the core principles of Machine Learning
- ▶ Connect theory to real business decision-making

Learning Outcomes

- ▶ Select appropriate models depending on the problem
- ▶ Interpret results and evaluate performance
- ▶ Understand limitations and trade-offs

Assessment & Practice

- ▶ Applied exercises throughout the course
- ▶ Mid-term exam (March 10th)
- ▶ Final group business project

Course Roadmap — Machine Learning for Business

Module 0 — Big Picture of Machine Learning

Module 1 — Probability and Statistical Foundations

Module 2 — Linear Models and Probabilistic Interpretation

Module 3 — Optimization

Module 4 — Generalization and Model Evaluation

Module 5 — Nonlinear Models and Ensembles

Module 6 — Machine Learning in Practice

Module 7 — Unsupervised Learning

Conclusion

Module 0 — Big Picture

- ▶ What is Machine Learning?
- ▶ Supervised vs Unsupervised Learning
- ▶ From Data to Decision-Making

What is Machine Learning?

Definition: Machine Learning (ML) enables algorithms to learn patterns and tasks directly from data without being explicitly programmed.

Key Points:

- ▶ ML lies at the intersection of **Mathematics** and **Computer Science**.
- ▶ A **Machine Learning Model** automates repetitive tasks.
- ▶ An **AI Agent** goes further: it takes decisions and actions based on triggers and observations.
- ▶ Two main strategies to teach a Machine Learning Model:
 - ▶ *Supervised Learning*: The model is guided using labeled data points (x_i, y_i) .
 - ▶ *Unsupervised Learning*: The model discovers structure or patterns in unlabeled data (x_i) .

Supervised Learning

Setup: We have input-output pairs:

$$(x_i, y_i), \quad i = 1, \dots, n$$

Goal: Learn a function $f(x)$ that predicts y accurately.

Two main types:

- ▶ *Regression:* Predict a continuous outcome (e.g., house prices, sales forecast)
- ▶ *Classification:* Predict a categorical outcome (e.g., loan default, churn yes/no)

Business Example: A bank wants to predict whether a new applicant will default (classification) or estimate the expected loan amount (regression).

Unsupervised Learning

Goal: Discover structure in unlabeled data.

We observe only features:

$$x_i, \quad i = 1, \dots, n$$

No target y is available.

Common Tasks:

- ▶ *Clustering*: Group similar data points (e.g., customer segmentation)
- ▶ *Dimensionality Reduction*: Summarize data with fewer variables (e.g., PCA for marketing analytics)

Business Example: A retail company wants to identify customer segments based on purchase behavior to design targeted marketing campaigns.

Data-Driven Approach

Idea: Use data to drive business decisions instead of intuition alone.

Steps:

1. Collect and clean relevant data
2. Explore and visualize patterns
3. Build predictive or descriptive models
4. Make decisions and take action
5. Monitor outcomes and iterate

Business Example: A bank wants to optimize credit offers:

- ▶ Analyze past customer data
- ▶ Predict likelihood of accepting a new offer
- ▶ Adjust campaigns dynamically

Module 1 — Probability Foundations

- ▶ Random Variables (Discrete and Continuous)
- ▶ Probability Distributions
- ▶ Expectation and Variance
- ▶ Normal Distribution
- ▶ Law of Large Numbers
- ▶ Central Limit Theorem
- ▶ Confidence Intervals
- ▶ Bayes' Theorem

Why Probability in Business?

Business Example: Loan Default

A bank evaluates loan applications.

- ▶ 8% of all customers default
- ▶ 25% of customers with low credit score default

Question:

What is the probability that a new applicant will default?

Key Idea: Business decisions are made under uncertainty.

Discrete Random Variables

Business Example: Online Shop

Let

X = Number of purchases per customer per month

Possible values:

$$X \in \{0, 1, 2, 3, \dots\}$$

Definition:

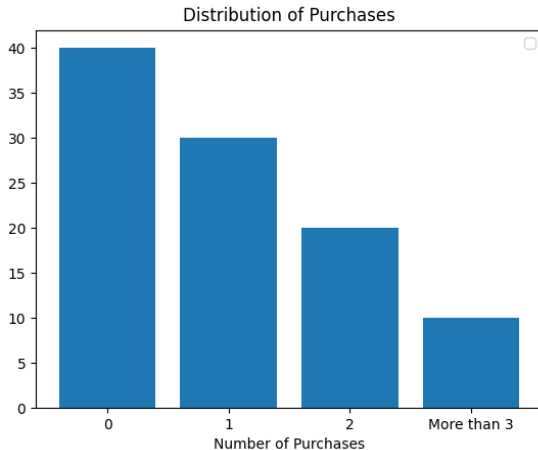
A *discrete* random variable is a function

$$X : \Omega \rightarrow \mathbb{N}$$

that assigns a numerical value to each outcome. An element $\omega \in \Omega$ is called an **observation** (here the behaviour of a customer).

Example: Probability Distribution of Purchases

Histogram for Discrete Distribution:



Discrete Probability Distribution

Business Example: Purchase Behavior

Purchases	Relative Frequency
0	0.4
1	0.3
2	0.2
3+	0.1

Definition:

For a discrete random variable:

$$\begin{aligned}p(x) &= \mathbf{P}(X = x) = \mathbf{P}(\{\omega \in \Omega | X(\omega) = x\}) \\&= \mathbf{P}(\{\text{all customers that buy } x \text{ times}\})\end{aligned}$$

with

$$\sum_x p(x) = 1$$

Relative Frequency

From Data to Probability

In practice, probabilities are often estimated from observed data.

If we observe n customers and n_x of them buy x times, the **relative frequency** is

$$f(x) = \frac{n_x}{n}$$

As the number of observations increases,

$$f(x) \rightarrow P(X = x)$$

Idea: Relative frequencies approximate probabilities.

Expectation of Discrete Probability Distribution

Business Example: Average Purchases

$$\mathbb{E}[X] = \sum_x x \cdot \mathbf{P}(X = x)$$

$$= 0(0.4) + 1(0.3) + 2(0.2) + 3(0.1) = 0.7$$

Interpretation:

- ▶ The purchase average is nearly 1
- ▶ It is possible to estimate the expected revenue

Continuous Random Variables

Business Example: Delivery Time

Let

X = Delivery time (in days)

Possible values:

2.13, 3.47, 4.002, ...

Definition:

A *continuous* random variable is a function

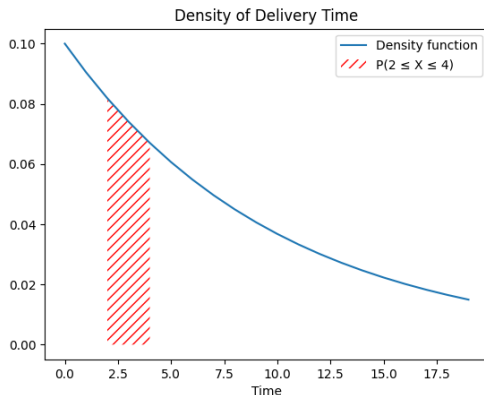
$$X : \Omega \rightarrow \mathbb{R}$$

that assigns a numerical value lying in the *uncountable* set of real numbers to each observation.

Example: Probability Density of Delivery Time

Assumption: The density function of delivery time follows an exponential distribution of parameter $\lambda = 0.10$ (i.e. $f(x) = \lambda \exp(-\lambda x)$ if $x \geq 0$, else 0).

Visualization of Probability:



Probability Density Function

Definition: A continuous random variable has a density $f(x)$ such that:

- ▶ $f(x) \geq 0$
- ▶ $\int_{-\infty}^{\infty} f(x) dx = 1$

Probability of an interval:

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f(x) dx$$

For continuous variables:

$$\mathbf{P}(X = a) = 0$$

Expectation of Probability Density Function

Business Example: Delivery Time

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f(x) dx = \frac{1}{\lambda} = 10$$

Business Interpretation:

- ▶ The average of delivery time is 10 days
- ▶ It is possible to estimate the delivery delay between two customers.

Variance and Risk

Business Question:

Two segments have the same average revenue. Which one is riskier?

Definition:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Variance measures variability and business risk.

Normal Distribution

Definition:

A random variable X follows a normal distribution if its density is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Notation:

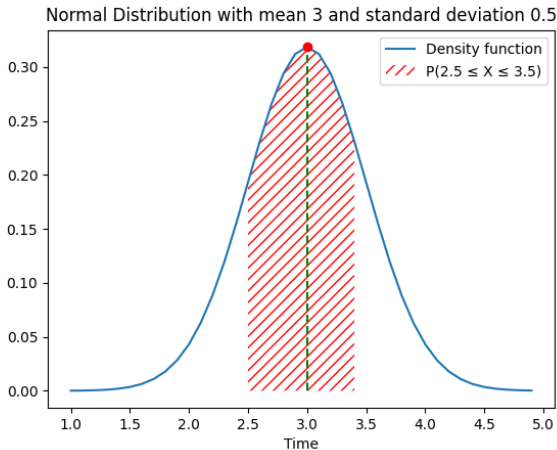
$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Properties:

- ▶ $\mathbb{E}(X) = \mu$: mean
- ▶ $\text{Var}(X) = \sigma^2$: variance

Normal Distribution Density

Illustration:



Example: Normal Delivery Time

Assume delivery time follows:

$$X \sim \mathcal{N}(3, 0.5^2)$$

- ▶ Mean = 3 days
- ▶ Standard deviation = 0.5 days

We can compute:

$$\mathbf{P}(2.5 \leq X \leq 3.5)$$

Interpretation of Parameters

Mean μ :

- ▶ Determines the center
- ▶ Shifts the distribution

Variance σ^2 :

- ▶ Controls spread
- ▶ Larger variance $\Rightarrow$ flatter curve
- ▶ Smaller variance $\Rightarrow$ more concentrated

The normal distribution is symmetric around μ .

The 68–95–99.7 Rule

Empirical Rule for the Normal Distribution

If

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

then approximately:

$$\mathbf{P}(|X - \mu| \leq \sigma) \approx 68\%$$

$$\mathbf{P}(|X - \mu| \leq 2\sigma) \approx 95\%$$

$$\mathbf{P}(|X - \mu| \leq 3\sigma) \approx 99.7\%$$

Business Interpretation:

Standard deviation directly quantifies risk and uncertainty.

Conditional Probability

Business Example: Online Advertising

$$\mathbf{P}(\text{Buy} \mid \text{Click})$$

Definition:

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}A \cap B}{\mathbf{P}(B)}$$

Conditional probability models dependence between events.

Fraud Detection Example

Insurance Fraud

- ▶ 2% of claims are fraudulent
- ▶ 90% of fraudulent claims are flagged
- ▶ 5% of legitimate claims are falsely flagged

Question:

If a claim is flagged, what is the probability it is truly fraudulent?

Bayes' Theorem

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(B \mid A)\mathbf{P}(A)}{\mathbf{P}(B)} = \frac{\mathbf{P}(B \mid A)\mathbf{P}(A)}{\mathbf{P}(B|A)\mathbf{P}(A) + \mathbf{P}(B|\overline{A})\mathbf{P}(\overline{A})}$$

Interpretation:

- ▶ $\mathbf{P}(A)$: Prior belief
- ▶ $\mathbf{P}(B \mid A)$: Likelihood
- ▶ $\mathbf{P}(A \mid B)$: Posterior belief

Bayes' theorem updates beliefs using new evidence.

Bayesian Update in Fraud Detection

$$\mathbf{P}(F \mid \text{Flag}) = \frac{0.9 \cdot 0.02}{0.9 \cdot 0.02 + 0.05 \cdot 0.98}$$

Result:

$$\mathbf{P}(F \mid \text{Flag}) \approx 27\%$$

Even highly accurate systems can produce many false alarms when the base rate is small.

A/B Testing Example

New Checkout Design

- ▶ After 20 users: conversion = 70%

A/B Testing Example

New Checkout Design

- ▶ After 20 users: conversion = 70%
- ▶ After 20,000 users: conversion = 12%

A/B Testing Example

New Checkout Design

- ▶ After 20 users: conversion = 70%
- ▶ After 20,000 users: conversion = 12%

Why does the estimate change so much?

Law of Large Numbers

Let

$$X_1, \dots, X_n$$

be n independent random variables with expectation μ .

Define the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Then:

$$\bar{X}_n \rightarrow \mu \quad \text{as } n \rightarrow \infty$$

Business Meaning of LLN

The Law of Large Numbers explains:

- ▶ Why large datasets reduce noise
- ▶ Why small samples are unreliable
- ▶ Why ML improves with more data

Large data stabilizes estimates. This comes from the following inequality which holds for all $\epsilon > 0$:

$$\mathbf{P}(|\bar{X}_n - \mu| > \epsilon) \leq \frac{V(\bar{X}_n)}{\epsilon^2}$$

Why Does the Average Look Normal?

Customer Spending Example

Individual spending is highly skewed.

Why Does the Average Look Normal?

Customer Spending Example

Individual spending is highly skewed.

However:

- ▶ The average of 500 customers
- ▶ Appears approximately normal

Why?

Central Limit Theorem

Let $X_1, \dots, X_n$ be i.i.d. with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2$$

Then:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \rightarrow \mathcal{N}(0, 1)$$

as $n \rightarrow \infty$.

Business Meaning of the CLT

For large n :

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Implications:

- ▶ Enables confidence intervals
- ▶ Enables A/B testing
- ▶ Uncertainty decreases with $\sqrt{n}$

Market Research Example

A survey of 1000 customers:

- ▶ 180 would buy the product

Estimated probability:

$$\hat{p} = 0.18$$

Is this the true purchase probability?

Population vs Sample

Population parameter:

p = True purchase probability

Sample estimate:

$$\hat{p} = \frac{\text{Number of buyers}}{n}$$

In practice, we only observe samples.

Confidence Interval

Using the CLT:

$$\bar{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

Interpretation:

If we repeated the study many times, 95% of the intervals would contain the true parameter.

Exercise: Histogram with Relative Frequency

- ▶ Is the wage distribution symmetric, left-skewed, or right-skewed?
- ▶ Where are most wages concentrated?
- ▶ Are there extreme values or outliers?

Exercise: Histogram with Relative Frequency

```
import pandas as pd
import matplotlib.pyplot as plt

# Load data
df = pd.read_csv("Wage_merged.csv")

wages = df["wage"]

# Histogram with relative frequency
plt.hist(wages, bins=20, density=True)

plt.xlabel("Hourly Wage")
plt.ylabel("Relative Frequency")
plt.title("Distribution of Hourly Wages")

plt.show()
```

Exercise: Law of Large Numbers (LLN)

- ▶ What happens to the sample mean when the sample size increases?
- ▶ What happens to the variance of the estimator as the sample size grows?
- ▶ Are there extreme values or outliers?

Exercise: Law of Large Numbers (LLN)

```
import numpy as np

wages = df["wage"]
true_mean = wages.mean()

sample_sizes = range(5, 500)
means = []

for n in sample_sizes:
    sample = wages.sample(n, replace=True)
    means.append(sample.mean())

plt.plot(sample_sizes, means, label="Sample Mean")
plt.axhline(true_mean, linestyle="--", label="True Mean")

plt.xlabel("Sample Size")
plt.ylabel("Mean Wage")
plt.title("Law of Large Numbers")

plt.legend()
plt.show()
```

Exercise: Central Limit Theorem (CLT)

- ▶ Is the sample means distribution symmetric, left-skewed, or right-skewed?
- ▶ What happens if we increase the sample size from 30 to 100?

Exercise: Central Limit Theorem (CLT)

```
sample_size = 30
num_samples = 1000

sample_means = []

for _ in range(num_samples):
    sample = wages.sample(sample_size, replace=True)
    sample_means.append(sample.mean())

plt.hist(sample_means, bins=30, density=True)

plt.xlabel("Sample Mean Wage")
plt.ylabel("Density")
plt.title("Central Limit Theorem")

plt.show()
```

Exercise: Confidence Interval for the Mean Wage

- ▶ If we collect new wages different from the dataset, what would be the range of values with 95% of confidence ?

Exercise: Confidence Interval for the Mean Wage

```
wages = df["wage"]

n = len(wages)
mean_wage = wages.mean()
std_wage = wages.std(ddof=1)

# Standard error
se = std_wage / np.sqrt(n)

# 95% confidence interval
ci = stats.t.interval(0.95, df=n-1, loc=mean_wage, scale=se)

print("Mean wage:", mean_wage)
print("95% confidence interval:", ci)
```

Exercise: One-Sample Hypothesis Test

- ▶ If we collect new wages different from the dataset, what would be the range of values with 95% of confidence ?

Exercise: One-Sample Hypothesis Test

```
from scipy.stats import ttest_1samp

t_stat, p_value = ttest_1samp(wages, 10)

print("t-statistic:", t_stat)
print("p-value:", p_value)
```

Exercise: Comparing Two Groups (Two-Sample Test)

```
men = df[df["female"] == 0]["wage"]
women = df[df["female"] == 1]["wage"]

from scipy.stats import ttest_ind

t_stat, p_value = ttest_ind(men, women)

print("t-statistic:", t_stat)
print("p-value:", p_value)
```

Exercise: Comparing Two Groups (Two-Sample Test)

- ▶ What is the null hypothesis in this test?
- ▶ What does the p-value tell us about the difference between the groups?

Module 2 — Linear Models

- ▶ Linear Regression (One Variable)
- ▶ Multiple Linear Regression
- ▶ Multivariate Random Variables
- ▶ Multivariate Gaussian Distribution
- ▶ Likelihood and Loss Functions
- ▶ Logistic Regression for Classification

Linear Regression (1 Variable)

Model:

$$Y = w_0 + w_1 X + \varepsilon$$

Assume:

$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Loss function (Least Squares):

$$L(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - w_0 - w_1 x_i)^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \approx \mathbb{E}[\|Y - \hat{Y}\|_2]$$

for n large

Multiple Linear Regression

Model:

$$Y = \mathbf{w}^T \mathbf{x} + \varepsilon = \mathbf{w}^T (\mathbf{x}_1, \dots, \mathbf{x}_n) + \epsilon$$

Assume:

$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Loss function:

$$L(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{w}^T \mathbf{x}_i)^2$$

Multivariate Random Variables

Business Example: House Features

$$\mathbf{x} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} \text{Size} \\ \text{Number of rooms} \end{pmatrix}$$

Definition:

A multivariate random variable is a vector:

$$\mathbf{x} = \begin{pmatrix} X_1 \\ \vdots \\ X_d \end{pmatrix}, \text{ where } X_i \text{ is a random variable}$$

Mean Vector and Covariance Matrix

Mean vector:

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_d] \end{pmatrix}$$

Covariance matrix:

$$\Sigma = \mathbb{E} \left[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T \right]$$

For two variables:

$$\Sigma = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \text{Var}(X_2) \end{pmatrix}, \text{ with}$$

$$\text{Cov}(X_1, X_2) = \mathbb{E}[(X_1 - \mathbb{E}[X_1])(X_2 - \mathbb{E}[X_2])]$$

Multivariate Gaussian Distribution

Definition:

A random vector $\mathbf{X} \in \mathbb{R}^d$ follows a multivariate normal distribution if:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

Notation:

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

Interpretation of the Covariance Matrix

- ▶ μ : center
- ▶ Σ : spread and dependence
- ▶ If $\text{Cov}(X_1, X_2) = 0$, ellipses aligned with axes
- ▶ If $\text{Cov}(X_1, X_2) \neq 0$, ellipses are tilted

Loss Functions Come from Probability Models

Model	Noise Assumption	Loss
Linear Regression	Gaussian	Squared Loss
Logistic Regression	Bernoulli	Cross-Entropy

Machine learning models are often derived from probabilistic assumptions.

Logistic Regression

Model:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-w^T x}}$$

Loss (Cross-Entropy):

$$L(w) = -\frac{1}{n} \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

Module 3 — Optimization

- ▶ General Optimization Problem
- ▶ Loss Minimization
- ▶ Gradient Descent
- ▶ Learning Rate
- ▶ Convex vs Non-Convex Problems
- ▶ Optimization Variants (SGD, Mini-batch)

Business Example: Training a Pricing Model

We want to predict house prices using linear regression.

Loss function:

$$L(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w^T x_i)^2$$

Goal:

$$\min_w L(w)$$

Question:

- ▶ How do we compute the best parameter w ?
- ▶ What if no closed-form solution exists?

General Optimization Problem

In machine learning, training a model means solving:

$$\min_{\theta} L(\theta)$$

- ▶ θ = model parameters
- ▶ $L(\theta)$ = loss function

Examples:

- ▶ Linear regression $\rightarrow$ squared loss
- ▶ Logistic regression $\rightarrow$ cross-entropy
- ▶ Neural networks $\rightarrow$ cross-entropy

Idea of Gradient Descent

Interpret the loss function as a landscape.

Goal: reach the lowest point.

The gradient:

$$\nabla L(\theta)$$

points in the direction of steepest increase.

To minimize the loss:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

► η = learning rate

Gradient Descent

We solve:

$$\min_{\theta} L(\theta)$$

Update rule:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

- ▶ η = learning rate
- ▶ $\nabla L(\theta)$ = gradient
- ▶ Repeat until convergence

Example: Gradient for Linear Regression

Loss:

$$L(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w^T x_i)^2$$

Gradient:

$$\nabla L(w) = -\frac{2}{n} X^T (y - Xw)$$

Update rule:

$$w \leftarrow w + \frac{2\eta}{n} X^T (y - Xw)$$

Learning Rate

Gradient descent update:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

Choice of learning rate η :

- ▶ Too small $\rightarrow$ slow convergence
- ▶ Too large $\rightarrow$ divergence or oscillation
- ▶ Well chosen $\rightarrow$ fast and stable convergence

Learning rate is a key hyperparameter.

Variants of Gradient Descent

Classical gradient descent:

- ▶ Uses entire dataset at each iteration

In large datasets, we use:

- ▶ Stochastic Gradient Descent (SGD)
- ▶ Mini-batch Gradient Descent

These methods are standard in deep learning.

Module 4 — Generalization

- ▶ Overfitting vs Underfitting
- ▶ Bias–Variance Tradeoff
- ▶ Regularization (L1 and L2)
- ▶ Decision Boundaries
- ▶ Performance Metrics:
 - ▶ R^2
 - ▶ Accuracy
 - ▶ Precision
 - ▶ Recall
 - ▶ F1-score

Overfitting vs Underfitting

Problem: How well does a model generalize to unseen data?

- ▶ **Underfitting:** Model is too simple
 - ▶ Fails to capture patterns in training data
 - ▶ High bias, low variance
 - ▶ Poor performance on both training and test sets
 - ▶ Example: Linear model trying to fit a quadratic trend
- ▶ **Overfitting:** Model is too complex
 - ▶ Captures noise as if it were signal
 - ▶ Low bias, high variance
 - ▶ Excellent performance on training set, poor on test set
 - ▶ Example: High-degree polynomial for limited data points

Business Example: A pricing model fits historical sales perfectly (overfit) but predicts future sales poorly.

Bias–Variance Tradeoff

Problem: Why does increasing model complexity sometimes hurt performance?

Prediction error can be decomposed into:

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

- ▶ **Bias:** Error from overly simple assumptions High bias → underfitting
- ▶ **Variance:** Sensitivity to small fluctuations in training data High variance → overfitting
- ▶ **Noise:** Irreducible randomness in the data

Business Example: A very simple demand model ignores seasonality (high bias). A very complex model reacts to random weekly fluctuations (high variance).

Regularization

Goal: Prevent overfitting by controlling model complexity.

General form of regularized loss:

$$L(\mathbf{w}) = \sum_{i=1}^n (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \lambda \Omega(\mathbf{w})$$

- ▶ λ controls the strength of the penalty
- ▶ Larger $\lambda \rightarrow$ simpler model

Method	Penalty	Effect
Ridge (L2)	$\sum w_j^2$	Shrinks coefficients
Lasso (L1)	$\sum w_j $	Produces sparse model

Business Example: In marketing analytics with 100 features, regularization avoids unstable and extreme coefficient estimates.

Decision Boundary in Logistic Regression

Goal: Understand how logistic regression separates classes.

Prediction rule:

$$Y = 1 \quad \text{if} \quad \sigma(w^T x) > 0.5$$

This is equivalent to:

$$w^T x = 0$$

- ▶ The equation defines a **hyperplane**
- ▶ Points on one side $\rightarrow$ Class 1
- ▶ Points on the other side $\rightarrow$ Class 0

Business Example: In credit scoring, the decision boundary separates high-risk from low-risk applicants based on their features.

Regression Metrics

- ▶ **R^2 (Coefficient of Determination):**

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Indicates proportion of variance explained by the model.
 $R^2 \approx 1$ is good.

- ▶ **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Penalizes large errors more heavily.

- ▶ **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Robust to outliers; measures average absolute deviation.

Classification Metrics - Part 1

- ▶ **Accuracy:** Fraction of correct predictions

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Simple overall measure, but can be misleading for imbalanced classes.

- ▶ **Precision:** Fraction of positive predictions that are correct

$$\text{Precision} = \frac{TP}{TP + FP}$$

Important when false positives are costly.

Classification Metrics - Part 2

- ▶ **Recall (Sensitivity):** Fraction of actual positives correctly identified

$$\text{Recall} = \frac{TP}{TP + FN}$$

Important when missing a positive is costly.

- ▶ **F1-score:** Harmonic mean of Precision and Recall

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Balances precision and recall.

Module 5 — Nonlinear Models

- ▶ Decision Trees
- ▶ Random Forest
- ▶ Boosting Methods
- ▶ Introduction to Neural Networks

Decision Trees

Principle

- ▶ Partition the feature space into regions
- ▶ Use threshold rules: $x_j < t$
- ▶ Produce piecewise constant predictions

Split Criteria

Classification:

$$\text{Gini} = 1 - \sum_k p_k^2$$

Regression:

Minimize squared error

Flexible model with low bias but high variance.

Instability of Decision Trees

Observation

- ▶ Small change in data → very different tree
- ▶ Structure of splits can change drastically

Consequence

- ▶ High variance model
- ▶ Prone to overfitting

Can we reduce variance while keeping flexibility?

Random Forest: Main Idea

Instead of one tree:

Build many trees

Each tree

- ▶ Trained on bootstrap sample of data
- ▶ Uses random subset of features at each split

Aggregation

- ▶ Regression → average prediction
- ▶ Classification → majority vote

Why Random Forest Works

Each individual tree:

- ▶ Low bias
- ▶ High variance

Averaging many de-correlated trees:

Reduces variance

Ensemble of unstable models $\rightarrow$ Stable predictor

Random Forest and Bias–Variance

Compared to a single tree:

- ▶ Similar bias
- ▶ Much lower variance

Random Forest:

Improves generalization performance

Often works well with minimal tuning.

Motivation for Boosting

Suppose a small tree is weak:

- ▶ Slightly better than random
- ▶ High bias

Key question:

Can many weak learners create a strong model?

Boosting: Main Idea

Core mechanism

Train models sequentially

Each new tree:

- ▶ Focuses on previous errors
- ▶ Corrects residual mistakes

Final model:

$$F(x) = \sum_{m=1}^M \gamma_m T_m(x)$$

Boosting Intuition

Step 1:

- ▶ Fit a small tree

Step 2:

- ▶ Identify errors
- ▶ Increase focus on difficult observations

Step 3:

- ▶ Fit new tree to residual errors

Repeat until performance stabilizes.

Boosting as Gradient Descent

Boosting can be interpreted as:

Gradient descent in function space

At each iteration:

- ▶ Compute residuals (negative gradient)
- ▶ Fit tree to residuals
- ▶ Update model

Examples: Gradient Boosting, XGBoost, LightGBM

Random Forest vs Boosting

Random Forest

- ▶ Trees trained independently
- ▶ Mainly reduces variance
- ▶ Easily parallelizable

Boosting

- ▶ Trees trained sequentially
- ▶ Mainly reduces bias
- ▶ Often higher predictive performance

Introduction to Neural Networks

One hidden layer network:

$$h = \sigma(W_1x + b_1)$$

$$\hat{y} = W_2h + b_2$$

Key idea

- ▶ Linear transformations
- ▶ Followed by nonlinear activation

Universal function approximator when sufficiently large.

Module 6 — ML in Practice

- ▶ Feature Engineering
- ▶ Train / Validation / Test Split
- ▶ Cross-Validation
- ▶ Practical ML Pipeline
- ▶ Model Monitoring
- ▶ Hypothesis Testing for Model Drift

Business Example: Raw Data Is Not Enough

Suppose we predict house prices.

Available variables:

- ▶ Size (m^2)
- ▶ Year built
- ▶ Zip code

Question:

Should we use them directly?

Feature Engineering

Feature engineering:

Transform raw data into informative variables

Examples:

- ▶ Age of house = Current year - Year built
- ▶ One-hot encoding of zip code
- ▶ Interaction: Size $\times$ Location
- ▶ Log-transform price

Better features $\rightarrow$ Better models

Common Feature Transformations

- ▶ Scaling (standardization)

$$x' = \frac{x - \mu}{\sigma}$$

- ▶ Log-transform for skewed variables
- ▶ Polynomial features
- ▶ Categorical encoding (one-hot)

Important for:

- ▶ Linear models
- ▶ Regularization
- ▶ Gradient descent

Why Train/Test Split?

Suppose our model achieves:

$$R^2 = 0.95$$

on training data.

Question:

Will it perform equally well on new customers?

Train / Validation / Test Split

Dataset is divided into:

- ▶ Training set → Fit model
- ▶ Validation set → Tune hyperparameters
- ▶ Test set → Final unbiased evaluation

Important:

Test set must not influence model choice

Cross-Validation

Instead of one split:

Use multiple splits

K-fold cross-validation:

- ▶ Split data into K folds
- ▶ Train on K-1 folds
- ▶ Validate on remaining fold
- ▶ Repeat K times

Provides more stable performance estimate.

Business Problem: Model Drift

Suppose churn model was deployed 6 months ago.

Question:

Has customer behavior changed?

If distribution changes:

$$P_{\text{train}}(X) \neq P_{\text{production}}(X)$$

Model performance may degrade.

Monitoring with Hypothesis Testing

We test:

H_0 : Distribution unchanged

H_1 : Distribution changed

Example tests:

- ▶ t-test (mean change)
- ▶ Kolmogorov–Smirnov test
- ▶ Chi-square test (categorical)

Monitoring Model Performance

We monitor metrics over time:

- ▶ Accuracy
- ▶ Precision / Recall
- ▶ R^2

Hypothesis test:

H_0 : Performance unchanged

If rejected:

- ▶ Retrain model
- ▶ Update features
- ▶ Re-evaluate pipeline

Typical ML Pipeline

1. Data collection
2. Feature engineering
3. Train/Validation split
4. Model training
5. Hyperparameter tuning
6. Final evaluation on test set
7. Deployment

Machine learning is an iterative process.

Module 7 — Clustering

- ▶ Introduction to Clustering
- ▶ K-Means Algorithm
- ▶ Importance of Distance Metrics
- ▶ Silhouette Score

Business Example: Customer Segmentation

Suppose we have customer data:

- ▶ Annual income
- ▶ Spending score

Goal:

Group customers into segments

No labels are available.

What is Clustering?

Clustering is an unsupervised learning task.

Given data points:

$$x_1, \dots, x_n$$

Partition them into K groups such that:

- ▶ Points in same cluster are similar
- ▶ Points in different clusters are dissimilar

K-Means Algorithm

Choose number of clusters K .

Goal:

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

Where:

$$\mu_k = \text{mean of cluster } C_k$$

K-Means: Algorithm Steps

1. Initialize K centroids randomly
2. Assign each point to closest centroid
3. Update centroids (compute cluster means)
4. Repeat until convergence

Convergence:

- ▶ Assignments stop changing
- ▶ Centroids stabilize

Importance of Distance

K-means relies on distance:

$$\|x_i - \mu_k\|$$

Typically:

Euclidean distance

Implications:

- ▶ Features must be scaled
- ▶ Sensitive to outliers
- ▶ Assumes spherical clusters

Why Feature Scaling is Crucial

Suppose we cluster customers using:

- ▶ Income (0–100,000)
- ▶ Spending score (0–100)

Without scaling:

Income dominates distance

Solution:

$$x' = \frac{x - \mu}{\sigma}$$

Choosing the Number of Clusters

Key question:

What is the optimal K ?

Common methods:

- ▶ Elbow method
- ▶ Silhouette score

Silhouette Score

For each point:

- ▶ a = average distance to points in same cluster
- ▶ b = smallest average distance to other clusters

Silhouette value:

$$s = \frac{b - a}{\max(a, b)}$$

Interpretation of Silhouette Score

$$-1 \leq s \leq 1$$

- ▶ $s \approx 1 \rightarrow$ well clustered
- ▶ $s \approx 0 \rightarrow$ overlapping clusters
- ▶ $s < 0 \rightarrow$ likely misclassified

Overall quality:

Average silhouette score

Business Interpretation

High silhouette score:

- ▶ Clear customer segments
- ▶ Targeted marketing possible

Low silhouette score:

- ▶ Segments overlap
- ▶ Clustering may not be meaningful

Clustering must be both:

Mathematically sound and business relevant

Conclusion

- ▶ From Probability to Business Decisions
- ▶ Choosing the Right Model
- ▶ The Complete ML Pipeline
- ▶ Responsible and Monitored Deployment